This often confuses many new Elixir developers. OTP does have "Telecom" in the name, but it has nothing to do with phone systems. It's simply a relic of the development history of Erlang and its corporate relationship with Ericsson.

OTP consists of Erlang libraries, the Erlang runtime system, ready-to-use components, and a set of design principles that can be followed by Erlang programs. While OTP is technically separate from Erlang, in practice they are always shipped as a unit. When you download Erlang, you'll download a package called "Erlang/OTP" which includes both the Erlang language and the OTP libraries.

The BEAM

Erlang, Elixir, and OTP don't really do much by themselves. They need a runtime to execute on top of. This is where the BEAM comes in. BEAM stands for "Bogdan/Björn's Erlang Abstract Machine," but everyone simply calls it "the BEAM."

We'll be taking a look at parallelism in Chapter 5, GenServers: Build Cities, Not Skyscrapers, on page 65, but the important thing for now is that the BEAM natively supports the primitives that allow Elixir to work like it does. If you took Elixir code and tried to run it on a runtime that didn't work the same way, a lot of the strengths would weaken or even disappear. The combination of Elixir and the BEAM is what makes it world-class.

The BEAM has evolved and changed over time. Its name has also changed, so who knows if we'll see a totally different iteration of it at some point in the future! There's an excellent timeline[12] that goes over the history of Erlang and the various runtimes. This may not be interesting to everyone, but it's a fascinating collection of the evolution of Erlang over time.

Put It All Together

It seems like a lot, but you need to know only a few things to get started:

- Elixir is the programming language that you'll directly work with.
- Erlang is a language that you likely won't need to worry about.
- OTP is a set of libraries that Elixir uses to provide critical concepts and features.
- The BEAM is what runs your code.

12. https://www.erlang.org/blog/beam-compiler-history/